European Commission

Infrastructures for AI in Science

Mutual Learning Exercise on National Policies for AI in Science

Second thematic report

PSF CHALLENGE

HORIZON EUROPE
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Report



Research and Innovation

Infrastructures for AI in Science. Mutual Learning Exercise on National Policies AI in Science

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EUROPEAN COMMISSION

Infrastructures for AI in Science

Mutual Learning Exercise on National Policies for AI in Science

Second thematic report

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Table of Contents

1. Introduction 5

1.1. AI in Science: background 5
1.2. Scope and purpose of the report 6
1.3. Methodology: Survey and Meeting 6

2. High Performance Computing 7

2.1. Background 8
2.2. Strategic Context 8
2.3. Survey Inputs 9
2.4. Pisa Meeting Inputs 9
2.5. Key Challenges 10
2.6. Policy Recommendations 11

3. Data 12

3.1. Background 12
3.2. Survey Inputs 13
3.3. Pisa Meeting Inputs 14
3.4. Key Challenges 16
3.5. Policy Recommendations 16

4. Foundation Models 17

4.1. Background 17
4.2. Survey Inputs 18
4.3. Pisa Meeting Inputs 18
4.4. Key Challenges 18
4.5. Policy Recommendations 19

5. AI Scientists 20

5.1. Background 20
5.2. Survey Results 21
5.3. Pisa Meeting Results 21
5.4. Key Challenges 22

5.5. Policy Recommendations 23

6. Conclusion 24

References 25

List of Tables

Table 1: Examples Data sharing initiatives 14

1. Introduction

> ***''AI won't replace the scientist, but scientists who use AI will replace those who don't''***
>
> ***Adapted from a Microsoft report, 'The Future Computed' (G4)***

This thematic report provides an overview of the current state and main policy challenges in relation to artificial intelligence (AI) infrastructure in the countries taking part in the Mutual Learning Exercise (MLE) for AI in Science. It is based on a survey of AI infrastructure across MLE participating countries, desk research, a country visit to Pisa which took place on 21 and 22 November 2024, and Ross D. King's expert views.

1.1. AI in Science: background

AI in science is the application of AI to science. It covers everything from individual scientists using traditional AI or machine learning (ML) techniques, deep learning, large language models (LLMs) and related foundation models, to fully autonomous "AI scientists" that can conduct scientific research on their own.

AI is currently transforming the very nature of science, promising to make the scientific process more efficient and significantly help address the immense problems the world faces: climate change, pandemics, food insecurity, chronic diseases, etc. AI is making a significant impact across the sciences, leaving few fields untouched, from materials science to drug design, from quantum physics to medicine. The utilisation of AI is becoming characteristic of early 21st century science[1].

The application of AI to science has traditionally been termed 'discovery science'[2]. The founding masterwork was the Dendral/Meta-Dendral project at Stanford in the 1960s and 70s[3]. The Dendral project produced the world's first expert system[4], while the Meta-Dendral project produced one of the first ML systems[5].

AI in science now typically requires data-intensive computing and ML to find representations and patterns in ever-increasing volumes of structured and unstructured data[6]. Over the past few years

[1] OECD (2023), *Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research*, OECD Publishing, Paris. https://doi.org/10.1787/a8d820bd-e; Stevens, R. (2023) Testimony of Rick Stevens, Associate Laboratory Director for Computing, Environment, and Life Sciences Argonne National Laboratory, to the Senate Energy & Natural Resources Committee: Hearing to examine recent advances in artificial intelligence and the Department of Energy's role in ensuring U.S. competitiveness and security in emerging technologies; Choudhary, A., Fox, G., Hey, T. (2023) Artificial Intelligence for Science: A Deep Learning Revolution. World Scientific Publishing Co Pte Ltd; DOE (2020), Opportunities and Challenges from Artificial Intelligence and Machine Learning for the Advancement of Science, Technology, and the Office of Science Missions, https://doi.org/10.2172/1734848 OSTI ID:1734848.

[2] Discovery science (also known as discover-based science) is a scientific methodology which aims to find new patterns, correlations, and form hypotheses through the analysis of large-scale experimental data. Langley, P.W., Herbert, A.S., Bradshaw, G. & Zytkow, J.M., (1987), Scientific Discovery Computational Explorations of the Creative Process, https://doi.org/10.7551/mitpress/6090.001.0001.

[3] Buchanan, B. G., Sutherland, G. L. & Feigenbaum, E. A. Heuristic, DENDRAL: A Program for Generating Explanatory Hypotheses in Organic Chemistry in Machine Intelligence Vol. 4 (eds Meltzer, B. & Michie, D.) 209–254 (Edinburgh Univ. Press, 1969); J. Lederberg, E. A. Feigenbaum, B. G. Buchanan, and R. K. Lindsay, Applications of Artificial Intelligence for Organic Chemistry: The DENDRAL Project. McGraw-Hill, 1980.

[4] Boden, M. (2006), Mind as Machine: A History of Cognitive Science, Oxford University Press.

[5] J. Lederberg, E. A. Feigenbaum, B. G. Buchanan, and R. K. Lindsay, Applications of Artificial Intelligence for Organic Chemistry: The DENDRAL Project. McGraw-Hill, 1980.

[6] Choudhary, A., Fox, G., Hey, T. (2023), Artificial Intelligence for Science: A Deep Learning Revolution. World Scientific Publishing Co Pte Ltd.

there has been a rapid increase in the amount of computing requirement for the training and deployment of advanced AI/ML systems[7]. The fundamental impact of AI in Science is to drive the economic cost of scientific research down: the intellectual part of science is now being done by humans *and* AI, while the experimental part is now done by humans *and* laboratory robotics.

The current and future importance of AI in science has been underlined by the 2024 awarding of the Nobel physics prize to AI pioneers of neural networks, and the Nobel chemistry prize for applications of AI to protein structure prediction.

AI is currently dominated by Big Tech companies in the US and China[8]. National governments cannot compete, nor can any single European company[9]. This is a fundamental strategic concern for Europe[10].

While the extent of AI's capabilities is not yet fully understood, it is already clear that there is potential for AI to vastly increase the speed and efficiency of scientific research. Science forms the basis of technology, and technology forms the foundation of our society. Internationally competitive technology is required for the health, economic and military security of Europe.

1.2. Scope and purpose of the report

AI infrastructure is essential for the development and deployment of AI in Science. This infrastructure includes computing resources, data access, storage and connectivity, security, scalability, ethical compliance, and cost management.

The thematic report comprises an analysis of national strategies and policies for fostering AI in science infrastructure, current challenges, and policy suggestions. It explores the state-of-the-art of the infrastructures for AI in science, examines the potential for innovations such as 'AI Scientists', and assesses Europe's competitiveness in this field[11].

1.3. Methodology: Survey and Meeting

To better understand the needs and challenges of AI for Science, a survey of AI infrastructures across participating countries and regions of the MLE, and a country visit to Pisa on 21 and 22 November 2024 were organised as part of the MLE. Ross D. King also conducted desk research.

[7] Sevilla, J., Heim, L., Ho, A., Besiroglu, T., Hobbhahn, M., & Villalobos, P. (2022, July), Compute trends across three eras of machine learning. In *2022 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE.

[8] European Commission: Directorate-General for Research and Innovation, Arranz, D., Bianchini, S., Di Girolamo, V. and Ravet, J. (2023), Trends in the use of AI in science – A bibliometric analysis, Publications Office of the European Union, https://data.europa.eu/doi/10.2777/418191; Schmallenbach, L., Bärnighausen, T.W. & Lerchenmueller, M.J. (2024), The global geography of artificial intelligence in life science research. Nat Commun 15, 7527 (2024). https://doi.org/10.1038/s41467-024-51714-x; Ahmed, N., Wahed, M., & Thompson, N. C. (2023), The growing influence of industry in AI research. *Science*, *379*(6635), 884-886. DOI: 10.1126/science.ade2420

[9] USA Congress (2024), Bipartisan House Task Force on Artificial Intelligence.

[10] Ibid.

[11] Directorate-General for Research and Innovation (2023), AI in Science: Harnessing the power of AI to accelerate discovery and foster innovation ISBN: 978-92-68-07765-8; Group of Chief Scientific Advisors, European Commission (2024), Successful and timely uptake of Artificial Intelligence in science in the EU ISBN 978-92-68-10084-4; Fredrik Heintz, Nabil Belbachir, and Edward Curry (2023), "Strategic Orientations Towards an AI, Data, Robotics Roadmap 2025-2027", Brussels – Adra; Fredrik Heintz, Nabil Belbachir, and Edward Curry, (2024), "Strategic Research, Innovation, and Deployment Agenda 2025-2027", Brussels – Adra; European Strategic Forum on Research Infrastructures - Landscape Analysis 2024;European Commission: Directorate-General for Research and Innovation, Arranz, D., Bianchini, S., Di Girolamo, V. and Ravet, J. (2023), *Trends in the use of AI in science – A bibliometric analysis*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/418191.

It was agreed to divide the subject area of infrastructures for AI in Science into three areas:

- High Performance Computing (HPC)
- Scientific Data
- Frontier AI infrastructure: Foundation models, and AI scientists.

In the survey, questions were sent to both country representatives responsible for policy, and to leading AI in Science scientists. 12 responses to the policy questions and 32 to the technical questions were provided by respondents. There were generally mature responses for questions related to HPC and Scientific Data, although many of the responses regarding HPC were from established HPC users rather than AI users. The responses regarding the questions about Frontier AI Infrastructure were much less mature or absent, despite the already clear future importance, of such infrastructure. This highlights the importance of the role of the MLE in sharing best practice. While these responses offer valuable insights, it is important to note that they are neither comprehensive nor representative of all EU countries or the entire scientific community. The limited sample size, combined with the possibility of self-selection bias, means some perspectives may be underrepresented. As such, the findings should be viewed as indicative rather than definitive and are intended primarily to stimulate further discussion and can guide more comprehensive research.

The country visit to Pisa on 21 and 22 November 2024 was attended by 40 participants. This included representatives from the MLE countries and regions (both policy and HPC expertise): from Italy, Flanders, Germany, Estonia, Ireland, the Netherlands, Norway, Romania, and Slovenia; scientific experts, policy officers from the European Commission; as well as invited experts from industry (e.g. Hugging Face), from the AI, Data and Robotics Association (ADRA), technical experts (HPC and data). Working sessions were organised on the current state-of-the-art in AI in Science, on HPC, on data, and advanced AI in Science infrastructure.

2. High Performance Computing

'AI is now a branch of HPC' Ian Foster



Picture 1: The Italian Super Computer Leonardo
Source: Leonardo CINECA

2.1. Background

By HPC, we mean large computing facilities of all kinds, traditional supercomputers, cloud-based clusters, graphics processor (GPU)-based systems for learning and inference, etc. (see, e.g., Figure 1 above).

Although historically separate, the convergence of HPC and AI has been driven by improvements of GPU hardware, which in turn has enabled breakthroughs in deep learning. This convergence is increasingly requiring vast computational and data resources. Such vast HPC resources are accessible only to large corporations and governments.

In the MLE all countries and regions were asked about their HPC resources. At the Pisa meeting there were representatives from the Juelich Supercomputing Centre, the Barcelona Supercomputing Centre, the Italian National HPC Centre, and SURF (NL).

The leading EU level support for HPC is EuroHPC[12]. It is a joint initiative, created in 2018, between the EU, European countries and private partners to develop a world-class supercomputing ecosystem in Europe. EuroHPC is a public-private partnership designed to advance HPC across Europe. It facilitates the pooling of resources from the European Union, participating EU Member States, associated states of the Horizon Europe and Digital Europe programmes, and private stakeholders. The EuroHPC Joint Undertaking (JU) operates with two primary objectives: to establish a pan-European supercomputing infrastructure and to support research and innovation in HPC. Headquartered in Luxembourg City, Luxembourg, the Joint Undertaking began operations in November 2018 under the oversight of the European Commission and gained autonomy in 2020.The EuroHPC JU is jointly funded by its members with a budget of around EUR 7 billion for the period 2021-2027[13]. However, the HPC capabilities of EuroHPC versus the US and China are significantly smaller[14].

SURF in the Netherlands is an example of best practice of integrating HPC access with support to scientists to adopt AI, and with IT services in general.

A current key initiative of the EuroHPC JU is the AI Factories call[15]. This is a substantial investment in HPC with the goal to bring together the key ingredients for success in AI: computing power, data and talent. The AI Factories initiative aims to create a thriving European ecosystem for training advanced AI models and developing AI solutions. The AI Factories will be interconnected and available to European AI startups, industry and researchers.

2.2. Strategic Context

Within the last few years, AI has been transformed by the development of LLMs such as ChatGPT. These have achieved breakthrough performance on a wide range of tasks that require human intelligence. Science does not currently have a good theoretical understanding of how LLMs work, instead their success has been driven by empirical engineering. Indeed, the success of LLMs has been very surprising to the AI community. It had been assumed by most people that achieving such high performance would require an explicit model of the world, explicit internal symbols, and a grounding in the world (i.e. a robot body). LLMs have none of these.

The success of LLM aligns with the 'scaling hypothesis'. This is the idea that the performance of ML models, particularly deep learning models such as neural networks, improves predictably as their size, the amount of training data, and computational resources are increased. This hypothesis has been proven correct over multiple orders of magnitude, and the question is now whether it will continue to work to produce new generations of ever smarter AI. Around a Trillion dollars are being invested in new HPC on the assumption that the scaling hypothesis will continue to work, mostly

[12] Petr Taborsky (2024), https://www.elise-ai.eu/sra-refresh/building-a-compute-infrastructure-for-european-ai.
[13] More information available at: Discover EuroHPC JU - EuroHPC JU (europa.eu).
[14] https://www.exascaleproject.org/
[15] AI Factories | Shaping Europe's digital future

by US and Chinese Deep Tech companies[16]. However, the hypothesis risks plateauing due to economic, technical, and environmental constraints. If the scaling hypothesis is wrong the investment of US and Chinese Deep Tech companies in vast new computers for AI may be as technically far-seeing as building the giant sundial at Jantar Mantar in Jaipur in the 18th century when mechanical clocks were clearly the way forward. In such a case future progress in AI will require advances in more efficient ML methods or hardware.

There are two important reasons to hope that the scaling hypothesis does not continue. The first is that Europe cannot financially compete in the brute-force development of AI through scaling. The second is that the share of world energy used by computing infrastructure is rising rapidly[17], and that building vast new HPC resources for scaling AI will make this significantly worse[18]. It is to be noted that Microsoft have recently invested in the notorious Three Mile Island Nuclear Site[19].

2.3. Survey Inputs

The main survey conclusion was that there is a disparity in HPC provision between the larger countries, such as Germany and Italy (where the senior scientists surveyed considered that HPC resources were adequate), and the smaller countries such as the Netherlands, Slovenia, and Romania, where respondents felt that they did not have adequate HPC infrastructure. This disparity reflects years of differences in investment.

Few countries had dedicated HPC facilities for AI. This is concerning because funding inertia and sociology suggest that HPC facilities will simply be used for their traditional applications, even if these applications are super-charged by AI. It was noticeable in the survey that most of the senior scientists surveyed came from the traditional HPC community rather than the AI community.

The relationship between capital costs and the running costs of HPC facilities seems to vary considerably between countries[20]. It would seem likely that this results in diverse qualities of service, but there is no evidence of this in the survey.

The sustainability of HPC investment is unclear given the recent rise in energy costs and stricter environmental regulations, including those in the EU following the 2019 Green Deal[21]. It is desirable that HPC facilities are placed close to sources of renewable energy. This is true for LUMI (Finland) but not for Jupiter (Germany).

2.4. Pisa Meeting Inputs

Participants emphasised the need to improve the user friendliness of HPCs and enhance their inter-compatibility and integration. In addition, they stressed the need to invest in related types of hardware such as laboratory equipment where data can be generated in a systematic and uniform fashion that can be more easily sharable. Such laboratories exist with an example being the Max Planck Institute in Germany, but they are not currently widespread in Europe (see Section 5).

It was noted that the process of accessing EuroHPC can be complicated depending on the type of request, with larger requests typically being more complex.

[16] USA Congress (2024), Bipartisan House Task Force on Artificial Intelligence.
[17] Jones, N. (2018), How to stop data centres from gobbling up the world's electricity. *nature*, *561*(7722), 163-166.
[18] Thompson, N. C., Greenewald, K., Lee, K., & Manso, G. F. (2020), The computational limits of deep learning. *arXiv preprint arXiv:2007.05558*, *10;* Benhari, A., Trystram, D., Dufossé, F., Denneulin, Y., & Desprez, F. (2024), Green HPC: An analysis of the domain based on Top500. *arXiv preprint arXiv:2403.17466*.
[19] Sherman, N. (2024), 'Microsoft chooses infamous nuclear site for AI power', BBC News, 20 September 2024, https://www.bbc.co.uk/news/articles/cx25v2d7zexo.
[20] OECD (2023), "A blueprint for building national compute capacity for artificial intelligence", *OECD Digital Economy Papers*, No. 350, OECD Publishing, Paris, https://doi.org/10.1787/876367e3-en.
[21] https://www.europarl.europa.eu/factsheets/en/sheet/71/environment-policy-general-principles-and-basic-framework

Considering the significant energy demands of HPC systems, participants suggested placing greater emphasis on developing AI systems that require less data in the future. This aligns with current experience, which shows that AI systems tailored to specific tasks can perform effectively even when built on smaller amounts of data.

Participants discussed that the EuroHPC JU Access for AI and Data-Intensive Applications call will take into account proposals for both AI for science and AI for industry, albeit the latter proposals have been encouraged more strongly, considering that historically AI for industry – including from start-ups – has not been developed and the European Commission would like to get this domain up to speed.

Participants recognised the need for harmonising education for HPC usage across Member States and exploring ways to transfer tacit knowledge, particularly from more advanced Member States towards those less advanced. Additional ideas included raising awareness in the Member States' public and private sector as regards the possibilities that HPC opens as well as harmonising access policies of HPCs facilities. For example, the LUMI super-computer in Finland makes available 50% of its computing resources to companies and scientists outside of the country, with companies, however, having to pay for usage, which is not the case for HPCs in other countries. It was noted that for HPC training EuroCC is very important. EuroCC is a European network of 33 national HPC competence centres. It provides a useful means for HPC training and cooperation amongst diverse stakeholders in Europe.

The meeting highlighted the need to unify software deployment in Europe and provide a set of standard software (e.g. through collaborative platforms) that can facilitate collaboration between scientists. It also highlighted that computational resources should be seamless for scientists, as currently, a lot of them still rely on computational resources on cloud.

2.5. Key Challenges

The following key challenges were identified. These are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- **Access to HPC:** Access to computing resources will be critical to scientific productivity[22], especially in some areas[23]. European scientists will require access to much greater HPC resources if Europe is to retain its competitiveness in AI in Science.
- **Disparities of Resources**: Scientists in under-resourced European countries, and in some application domains, face significant barriers to access HPC resources and expertise for AI in Science[24].
- **Industry Dominance**. Commercial firms dominate computing resources for AI in science, which makes it easier to hire talent, develop industry-leading AI benchmarks, and continue to invest in research[25].

[22] McKeon, J., & Boudou, J. (2024), Innovation Under Resource Constraints: Supercomputing in Scientific Research, *Available at SSRN 4868123*.
[23] Bianchini, S., Müller, M., & Pelletier, P. (2023), Drivers and Barriers of AI Adoption and Use in Scientific Research, *arXiv preprint arXiv:2312.09843*.
[24] Van Noorden, R., & Perkel, J. M. (2023). AI and science: What 1,600 researchers think, *Nature*, *621*(7980), 672-675.
[25] Ahmed, N., Wahed, M., & Thompson, N. C. (2023), The growing influence of industry in AI research, *Science*, *379*(6635), 884-886; Besiroglu, T., Bergerson, S. A., Michael, A., Heim, L., Luo, X., & Thompson, N. (2024), The Compute Divide in Machine Learning: A Threat to Academic Contribution and Scrutiny?. *arXiv preprint arXiv:2401.02452*.

- **Sustainability**: The environmental and financial sustainability of future scientific computing and HPC infrastructure is unclear[26]. European countries need harmonised practices and metrics for measuring the energy consumption of HPC systems[27].
- **Usability**: Algorithms are one of the most important parts of computing, alongside hardware and software[28]. European HPC systems need to become more user-friendly and integrated with data sources (see the example of SURF).
- **Educational Gaps**: Expert support is needed to deploy AI in science on HPC resources. There is a pressing need for interdisciplinary training to bridge AI and HPC knowledge[29].

2.6. Policy Recommendations

A number of key recommended actions for policymakers should be taken to address the key challenges highlighted above. These recommendations are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- The participating MLE countries should invest more in HPC resources. These resources need to be designed both for learning and for inference.
- MLE countries (regions) should engage with the AI Factories initiative to boost HPC availability. It is noteworthy that Flanders, Ireland, and the Netherlands have not yet been involved in any submitted bids.
- A top priority for European AI research should be the development of smaller, more energy efficient AI systems. Developing such models is essential for Europe to remain competitive in AI. More efficient AI might be achieved by improving the current Deep Learning paradigm, or through other approaches such as integration with symbolic AI methods. Therefore, these areas of research should be supported.
- European Structural and Investment Funds (ESIF) should be investigated as a source of funding to make access to HPC for AI in Science more equitable across the EU.
- Energy efficiency and sustainability must be prioritised in HPC practices to address rising environmental and economic challenges. Promoting the use of renewable energy for new HPC facilities and providing incentives for retrofitting existing centres would significantly reduce their environmental footprint.
- An inventory of the HPC software gaps and assets that are present at the national and European level is needed.
- The integration of AI and HPC into education and training programmes is critical to building a skilled workforce capable of leveraging these technologies effectively. Interdisciplinary Master's and PhD programmes should be introduced, blending AI and HPC knowledge to address emerging challenges. Furthermore, formal training initiatives must be established, and made accessible to researchers, industry professionals, and private sector participants, to foster a robust talent pipeline.
- Strengthening pan-European collaboration is essential for maximising the impact of HPC investments. Partnerships between MLE countries, and beyond, should be encouraged to

[26] Lannelongue, L., Aronson, H. E. G., Bateman, A., Birney, E., Caplan, T., Juckes, M., ... & Inouye, M. (2023), GREENER principles for environmentally sustainable computational science, *Nature Computational Science*, *3*(6), 514-521; Dhar, P. (2020), The carbon impact of artificial intelligence. *Nat. Mach. Intell.*, *2*(8), 423-425.
[27] https://ceec-coe.eu/wp-content/uploads/2024/08/ceec-bpg-measuring-energy.pdf
[28] Thompson, N. C., Ge, S., & Sherry, Y. M. (2021), Building the algorithm commons: Who discovered the algorithms that underpin computing in the modern enterprise?, *Global Strategy Journal*, *11*(1), 17-33.
[29] https://royalsociety.org/news-resources/projects/science-in-the-age-of-ai/ (see especially Ch.3)

share infrastructure, expertise, and best practices, fostering a culture of cooperation. Alignment of national and EU-level investments will prevent duplication, prioritise funding, and ensure resources are utilised efficiently to address shared priorities.

- Balancing collaboration between industry and academia is crucial to driving innovation while preserving open science principles. Pre-competitive partnerships should be supported to facilitate joint advancements in HPC and AI. Additionally, private companies should be incentivised to share data and infrastructure with the research community, creating a mutually beneficial ecosystem that leverages collective expertise and resources.

3. Data

3.1. Background

By data infrastructure for AI in Science, we mean data repositories, data sharing platforms, model sharing platforms (e.g. Hugging Face), etc[30].

It is important to note that recent advances in AI (LLMs) have been driven as much by access to vast amounts of data as by access to HPC. The central importance of data is also true in AI in science. For example, AlphaFold would have been impossible without open access to curated protein sequence and structure databases. These bioinformatics databases were publicly funded for decades.

Open Science is at the centre of European research policy[31]. Policies, initiatives and structures are developed and implemented to open up European science and research to make them more efficient and productive, seamless, transparent and robust as well as responsive to policy and society's needs and expectations. Notable Open Science practices include among others:

- early and open sharing of research:
  - pre-registration, registered reports, data deposition in shared repositories, pre-prints,
  - open collaboration within science and with other knowledge producers/users,
- providing immediate and unrestricted open access to scientific publications, research data, models, algorithms, software, protocols, notebooks, workflows, and all other research outputs,
- ensuring verifiability and reproducibility of research outputs,
- practicing responsible research output management (publications, data, and other outputs) in line with the FAIR (Findable, Accessible, Interoperable, and Reusable) principles[32].

Most of these practices directly address the provision of research data and its accessibility, interoperability, and trustworthiness. In particular, research data following the FAIR principles can be considered a key ingredient for AI in science.

A notable Open Science enabler in the field of research data and a key element of the European Open Science Policy is the European Open Science Cloud (EOSC). It is one of the Common European Data Spaces[33] and is therefore a main component of the European Strategy for Data[34] , enhancing the EU's leadership in the global data economy. In addition to incentivising the above

[30] EOSC Observatory: EOSC Observatory (eosc-portal.eu); European Strategic Forum on Research Infrastructures - Landscape Analysis 2024 landscape2024.esfri.eu

[31] https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/open-science_en

[32] Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., ... & Mons, B. (2016), The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data, 3(1), 1-9.

[33] https://digital-strategy.ec.europa.eu/en/policies/data-spaces

[34] https://digital-strategy.ec.europa.eu/en/policies/strategy-data

practices through projects carried out in the context of Horizon Europe throughout the last years, the first Node of the emerging EOSC Federation was launched by the European Commission in October 2024, the EOSC EU Node[35]. In the upcoming years the EOSC Federation is expected to connect more of these EOSC Nodes, building a web of distributed data infrastructures. In this context, many countries, regions or organisations follow similar principles as listed above, have corresponding policies in place and are about, or already have, existing research data infrastructures (see survey results) that could contribute to the EOSC Federation. A crucial element remains the usage of such infrastructures, i.e. to incentivise scientists to engage in Open Science practices. This, among others, includes training and skills development for researchers, an adequate research assessment rewarding data sharing and highlighting the added value of data sharing.

For EOSC and Open Science policies and practices in general the EOSC Steering Board[36] initiated a monitoring process, which since 2021 is carried out through yearly surveys among the Member States of the European Union and the Associated Countries to Horizon Europe. Its results are, on the one hand, published through the EOSC Open Science Observatory[37], but also in yearly reports[38] and a brochure highlighting use cases[39]. This brochure exemplifies the countries' efforts regarding corresponding data policies, data and service infrastructures or training.

3.2. Survey Inputs

The survey results suggest that most countries have national policies for data sharing in place and are aligned with EU data sharing initiatives. For example, the data policies of the two main funders in the Netherlands NWO and ZonMW are very similar to the data policies of the European Commission: they require that researchers submit a data management plan if they were awarded funding, and that they share research data resulting from the funded projects. Moreover, the requirements for the data management plan are aligned with the Core Requirements of Science Europe.

Most countries could also cite successful data-sharing initiatives. A few examples from the MLE survey are provided in the table below.

Country	Data sharing initiatives
Austria	• Austrian Data Portal: This portal provides access to a wide range of public data sets from various Austrian federal and regional authorities. It promotes transparency and encourages the use of open data in research and innovation[40]. • AUSSDA plays a crucial role in the Austrian research landscape by promoting data sharing and facilitating access to social science data, thereby enhancing research quality and collaboration in the field[41].
Ireland	• National Energy Research Database[42].

[35] https://open-science-cloud.ec.europa.eu/about/eosc-eu-node
[36] https://ec.europa.eu/transparency/expert-groups-register/screen/expert-groups/consult?lang=en&groupID=3756
[37] https://eoscobservatory.eosc-portal.eu/home
[38] E.g. for 2022: https://zenodo.org/records/10696674
[39] https://zenodo.org/records/10822698
[40] https://www.data.gv.at/
[41] https://aussda.at/
[42] https://www.seai.ie/seai-research/research-database

Country	Data sharing initiatives
	• National Open Research Forum[43]. • Ireland's Open Data Portal[44]. • Surveying Open Data Portal[45]. • Geospatial Data Hub[46].
Romania	• Medicai is a cloud-based medical imaging platform which is used by multiple hospitals and clinics across Romania. It facilitates collaboration among doctors, patients, and clinics. Medicai allows for the upload, sharing, and analysis of medical images (CT, MRI, X-rays) in a secure and efficient manner[47].
Slovenia	• The Oncochain platform, a private initiative, facilitates the collection, curation, and sharing of oncology data from hospitals and clinics. Oncochain ensures structured data exchange between healthcare providers for secondary use in clinical research and patient care. The platform demonstrates the potential for well-structured data sharing initiatives that align with healthcare needs while overcoming some challenges in data collection and management[48] . • Open Science Slovenia - a national Open Science Portal and repositories for open access to theses and research results of researchers at the Slovenian universities[49].

Table 1: Examples Data sharing initiatives

3.3. Pisa Meeting Inputs

It was noted that research data sharing also comes with certain restrictions, particularly depending on the domain area: applying AI to biomedical data raises privacy and ethical concerns; a more careful and balanced approach is required. In sensitive fields where privacy risks or potential dual-use issues exist, access to advanced models may need to be restricted, with closer oversight of their performance. Establishing clear governance structures, decision-making processes, and accountability for monitoring and reporting is therefore key for a successful ecosystem for data.

It was agreed that there is a need for more training and skills development for researchers. It is noteworthy that the sharing of data has been most successful in areas such as DNA sequences and protein structures, high-energy physics, astronomy, where the experimental context can be well-described by the research community. In other areas, such as laboratory biomedicine, the sharing of data is difficult because it requires a large amount of effort to describe the experimental context. A broad overview of researchers practices regarding data sharing is contained in the European Research Data Landscape[50] report.

[43] https://dri.ie/norf/
[44] https://data.gov.ie/
[45] https://data-osi.opendata.arcgis.com/
[46] https://www.geohive.ie/
[47] https://www.medicai.io/
[48] For more details, you can visit Oncochain's official website - https://www.oncochain.com/.
[49] https://www.openscience.si/default.aspx.
[50] https://op.europa.eu/en/publication-detail/-/publication/03b5562d-6a35-11ed-b14f-01aa75ed71a1/language-en

The meeting discussed the role of EOSC as a key enabler for digitalising research infrastructure and has the potential to be the most used infrastructure in Europe with the total investments in the cloud for the lifespan of Horizon Europe estimated at over EUR 1 billion. The Horizon Europe annual work programme in 2025 will include four distinct calls focusing on AI applications in the context of the EOSC.

The use by companies of publicly funded data, and whether they should be required to pay for its use, was discussed. The AI Act includes provisions on introducing accountability policies in general purpose AI-models and according to some participants, efforts are underway to make the process of analysis more transparent and showcase how data is used to build different models. However, once a model is built and (re)distributed, tracing the information which it used (e.g. the specific scientific articles) is generally not possible. In this context, a participant also noted that the value of keeping data open is much larger than the costs of running the data infrastructure and suggested that it would be advisable to continue with an open data model without having too much infrastructure to do tracking on top of that.

With regard to Open Science and data sharing, the participants advocated for Open Science and the enormous opportunities that come with it, although it was noted that there is still a challenge in using open data in downstream applications. As regards the administrative burden associated with using data, it was suggested that this can sometimes be excessive and discouraging for researchers, especially when it involves personal health data. In this respect, it was suggested that this administrative load contributes to Europe lagging behind other countries, where regulations are minimal or non-existent and advocated for a more streamlined approach without compromising essential regulatory requirements.

Philip Piatkiewicz, Secretary General of ADRA (AI Data Robotics Association) discussed three pillars necessary to integrate AI in the robotics process. The first pillar was the need to consider end-users and make sure that AI solutions are practical and relevant to them. The second pillar was fostering public-private partnerships and ensuring that the AI solutions are applicable to the real world. Finally, the third pillar was about providing incentives for data sharing.

A key issue that emerged in the discussion was a need to find ways to develop AI technology in line with the regulations that are in place based on the evidence base that is already available. Another widely discussed issue was the need to find ways to reward research staff making its data available (e.g. by uploading them on GitHub) given the high administrative burden and the time this task diverts from actual research. To address this, it was suggested that institutions integrate these activities into their career progression process. As one participant cited: "*Sometimes, I can see that there is a technology transfer that moves to companies, and this is just as valuable as getting a paper out [...] you have to architect the culture of recognising where the contributions are and awarding them accordingly."*

The meeting discussed the potential malevolent use of data and codes. Examples included the use of Llama by the Chinese military, and the use of fish migration data by poachers. To address such concerns, participants proposed implementing thorough risk assessments, control measures, and, where necessary, legal frameworks. They also emphasised that fear of misuse should not lead to overly restrictive data access or overshadow the benefits of open data and code, such as the rapid development of vaccines.

The concept of 'data stewards' was discussed. These could be an effective way to facilitate data sharing, alongside efforts for behavioural change amongst the research community. Such new roles and structures are already in place in some Member States. For example, the Netherlands has digital competence centres dealing with data sharing while Slovenia has a support centre for Open Science and training.

Finally, the MLE expert Stefano Bianchini discussed the challenges and opportunities that have to be solved in order to accelerate AI in science. In this regard, participants highlighted the need to (a) find out a good way not only to access but also to describe (meta)data and understand the biases that they might be introducing to models, (b) merge and connect different datasets that "do

not speak to each other" and (c) involve citizens more actively, and directly into the research process and create an overall inclusive research ecosystem.

3.4. Key Challenges

The following key challenges were identified, which are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- **Disparities in data infrastructure adoption and usage.** While initiatives like the EOSC aim to unify and enhance data and metadata sharing across Member States, adoption remains low. There are significant disparities in access to and use of infrastructure and data governance across regions[51].
- **Interoperability deficiency and siloed data.** Many datasets remain siloed and incompatible across and within disciplines, hindering collaboration and limiting the potential for AI to draw insights from integrated data[52]. A lack of standardised formats and harmonised protocols exacerbates this issue, impeding cross-disciplinary research.
- **Administrative and incentive barriers to data sharing.** Researchers face substantial administrative burdens when complying with data management requirements. This discourages engagement with Open Science practices[53]. Current incentive structures often fail to adequately reward researchers for data sharing (including maintenance and promotion of standards), limiting widespread participation[54]. This applies not only to 'data' in the conventional sense, but also to the algorithms, tools, and workflows that led to that data.
- **Ethical and legal concerns.** In areas like medicine and healthcare, privacy risks, and potential dual use concerns complicate data sharing[55].

3.5. Policy Recommendations

A number of key recommended actions for policymakers should be taken to address the key challenges highlighted above. These recommendations are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- Interoperability and cross-disciplinary collaboration should be prioritised to maximise the value of shared data. Standardising data formats and platforms will help break down silos, allowing researchers from diverse fields to work together effectively. Encouraging public-private partnerships will further enhance resource sharing and bring industry expertise to bear on scaling AI applications.

[51] Chen, R. (2021), *Mapping data governance legal frameworks around the world: Findings from the Global Data Regulation Diagnostic*. The World Bank.

[52] Vorisek, C. N., Klopfenstein, S. A. I., Löbe, M., Schmidt, C. O., Mayer, P. J., Golebiewski, M., & Thun, S. (2024), Towards an Interoperability Landscape for a National Research Data Infrastructure for Personal Health Data. *Scientific Data*, *11*(1), 772.

[53] Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., ... & Mons, B. (2016), The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, *3*(1), 1-9.

[54] Longpre, S., Mahari, R., Chen, A., Obeng-Marnu, N., Sileo, D., Brannon, W., ... & Hooker, S. (2024), A large-scale audit of dataset licensing and attribution in AI. *Nature Machine Intelligence*, *6*(8), 975-987.

[55] Mittal, S., Thakral, K., Singh, R., Vatsa, M., Glaser, T., Canton Ferrer, C., & Hassner, T. (2024), On responsible machine learning datasets emphasizing fairness, privacy and regulatory norms with examples in biometrics and healthcare. *Nature Machine Intelligence*, *6*(8), 936-949; Kaissis, G. A., Makowski, M. R., Rückert, D., & Braren, R. F. (2020), Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, *2*(6), 305-311.

- Improved training and the funding of dedicated support roles will be essential. Interdisciplinary educational programmes need to equip researchers with the skills needed for AI and data stewardship.
- Strengthening the incentives for data sharing should be considered, along with streamlined processes and reward systems that recognise researchers' contributions to open data repositories. By integrating these contributions into career evaluations and funding criteria, researchers could be encouraged to actively participate in Open Science initiatives. Such approaches have not worked very well in the past, but should still be tried.
- The EOSC should be expanded and standardised to realise its full potential. Aligning policies across Member States, enhancing the usability of centralised repositories, and implementing secure protocols for sensitive data sharing will create a seamless, collaborative environment. This will ensure equitable access to resources and promote widespread adoption.
- AI has the potential to extract explicit executable knowledge from textbooks, scientific papers and research outputs in general and as such enrich data infrastructures, e.g. by contributing to their knowledge commons and knowledge graphs. This area of AI research should be supported.

4. Foundation Models

4.1. Background

AI Foundation Models are a generalisation of LLMs. A Foundation Model is an AI model that is trained on broad data such that it can be adapted (e.g. fine-tuned) and applied across a wide range of applications[56]. The building of Foundation Models is typically very expensive, requiring access to large amounts of data and access to HPC. In contrast, adapting an existing Foundation Model to an application is much less expensive.

Foundation Models are set to revolutionise scientific research by allowing large, adaptable models to learn from vast datasets. Given the complexity of biological, chemical, and physical systems, this technology holds great potential for tackling key challenges in biology and medicine. Foundation Models display remarkable generalisation capabilities that make them highly versatile and easily adaptable to new tasks and problems, even without further training[57].

Foundation Models for science are a very promising way to provide the infrastructure to accelerate scientific research. For example, protein language models can be used to design new hypothetical structures with desired properties, which can then be experimentally tested[58]. Another successful multi-modal Foundation Model integrates gene expression with chromatin accessibility data[59]. Foundation Models can also be valuable to the social sciences, for example by enhancing survey research, online experiments, agent-based models, and other techniques commonly used to study human behavior[60].

[56] Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021), On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.
[57] Brown, T.B. et al. (2020), 'Language Models are Few-Shot Learners', arXiv [cs.CL], available at: http://arxiv.org/abs/2005.14165; Bommasani, R. et al. (2021), 'On the Opportunities and Risks of Foundation Models', available at: http://arxiv.org/abs/2108.07258 (Accessed: 26 March 2024).
[58] Watson, J.L. et al. (2023), 'De novo design of protein structure and function with RFdiffusion', Nature, 620(7976), pp. 1089–1100.
[59] Ashuach, T. et al. (2023), 'MultiVI: deep generative model for the integration of multimodal data', Nature methods, 20(8), p. 1222.
[60] Bail, C. A. (2024), Can Generative AI improve social science?. *Proceedings of the National Academy of Sciences*, *121*(21), e2314021121.

There is a need for public domain Foundation Models. A notable initiative is the Trillion Parameter Consortium: a new international consortium formed to create trustworthy and reliable generative AI models for science[61].

4.2. Survey Inputs

The survey demonstrated that there was agreement across almost all countries that Foundation Models are an important development in future science infrastructure. However, the survey highlighted conceptual and terminological confusion between the terms LLMs and Foundation Models. While it is true that LLMs are a form of Foundation Model, national efforts to train LLMs in Dutch, Swedish, or Finnish, are not immediately relevant to science.

4.3. Pisa Meeting Inputs

It was agreed that Foundation Models are increasingly central to research, especially when finetuned for specific downstream tasks, as they can create efficiencies (by using smaller label datasets which reduce overheads and are computationally relatively cheap). Making such Foundation Models available to researchers who might not have access to HPC presents some challenges (such as having access to compute where the model is located) but simultaneously has the potential to democratise the use of the models and help researchers answer specific research questions. It was noted that in contrast to the large investments needed to build infrastructure such as HPCs, the finetuning of the models can be done with different computational architecture that does not require investments of equivalent size.

It was noted that the public sector lags behind private companies in the field of AI – even when excluding Google and Microsoft which do not disclose information on these issues[62]. It was agreed that the main challenges associated with developing AI in science lies mostly in engineering, as Foundation Models are by an overwhelming extent an engineering task, whereas challenges related to copyright and licensing also exist.

The validation of Foundation Models was discussed. There are significant challenges involved in this process, as it requires extensive expertise and the design of tailored experiments. Despite these difficulties, it was emphasised that significant efforts to estimate prediction uncertainty and collaboration with experimental partners to validate the most compelling aspects of their findings have been made.

Finally, the ownership of Foundation Models was discussed. Some participants argued that the ownership of Foundation Models should be with the collaborators who own and understand this data best. Others thought the models should be more open and controlled by data science professionals.

4.4. Key Challenges

The following key challenges were identified, which are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- **Insufficient infrastructure for scaling and inference:** The future demands for learning and inference of Foundation Models, and the integration of these models into experimental workflows, requires significant new investment in infrastructure. In particular, expanding inference-specific capacity will be necessary - such as recently implemented at LUMI.

[61] Trillion Parameter Consortium (2024), https://www.anl.gov/article/new-international-consortium-formed-to-create-trustworthy-and-reliable-generative-ai-models-for

[62] USA Congress (2024), Bipartisan House Task Force on Artificial Intelligence.

- **Ethical and privacy concerns:** In fields involving sensitive data, such as healthcare and biomedicine, law or education privacy risks and potential dual-use scenarios necessitate stricter oversight. Balancing open data access with ethical considerations will require robust governance frameworks to monitor the capabilities and applications of Foundation Models while minimising misuse[63].
- **Public vs. private sector gaps in development and accessibility:** The public sector lags behind private companies in developing and deploying Foundation Models, due to engineering challenges, limited access to cutting-edge resources, and unresolved issues around copyright and licensing. These disparities hinder equitable access and collaboration, slowing scientific advancements reliant on these technologies. However, it is important to note that nearly two-thirds of the new Foundation Models released in 2023 (149 in total) were open-sourced, compared to one-third in 2021[64].
- **Trustworthy AI - black boxes vs. scientific knowledge:** Similarly to LLMs, Foundation Models have a tendency to hallucinate, and they cannot explain their output: they have no notion of truth, or even probability. This is unsatisfactory for scientific understanding and science in general[65]. The architecture of Foundation Models means that it is very hard to see how the hallucination problem can be solved, or correct logical reasoning guaranteed.
- **Illusion of understanding**. Foundation Models make scientists vulnerable to illusions of understanding and objectivity, and foster the growth of scientific monoculture[66].
- **Scientific misconduct and dilution of quality:** Foundation Models make it much easier to commit scientific misconduct. Scientific misconduct is the "*Intention[al] distortion of the research process by fabrication of data, text, hypothesis, or methods from another researcher's manuscript form or publication; or distortion of the research process in other ways*"[67].

4.5. Policy Recommendations

A number of key recommended actions for policymakers should be taken to address the key challenges highlighted above. These recommendations are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- Strategic investment in infrastructure (HPC & data) for Foundation Models is critical. Expanding HPC and data infrastructure to support both the training and inference of scientific Foundation Models is required. Future growth in data and computing demands necessitates the development of resources such as dedicated inference nodes.
- Interdisciplinary collaboration and targeted training programmes will help maximise the utility of Foundation Models. Educational initiatives are needed to bridge AI, HPC, and various scientific disciplines, creating a workforce skilled in leveraging these advanced tools. Collaborative research across institutions and countries should also be encouraged to integrate Foundation Models with hypothesis generation and experimental validation, driving breakthroughs across multiple fields of study.

[63] Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021), On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.
[64] https://aiindex.stanford.edu/report/#individual-chapters (see Chapter 1).
[65] Krenn, M., Pollice, R., Guo, S. Y., Aldeghi, M., Cervera-Lierta, A., Friederich, P., ... & Aspuru-Guzik, A. (2022), On scientific understanding with artificial intelligence. *Nature Reviews Physics*, *4*(12), 761-769.
[66] Messeri, L., & Crockett, M. J. (2024), Artificial intelligence and illusions of understanding in scientific research. *Nature*, *627*(8002), 49-58.
[67] Nylenna, M.; Andersen, D.; Dahlquist, G.; Sarvas, M.; Aakvaag, A. (1999), "Handling of scientific dishonesty in the Nordic countries. National Committees on Scientific Dishonesty in the Nordic Countries". Lancet. **354** (9172): 57–61.

- There is a need to equalise access for European scientists to Foundation Models. Open sharing of data and models should be promoted to ensure broad participation, enabling researchers across all sectors and regions to contribute to scientific advancements driven by AI.
- There is a need to develop general trustworthy Foundation Models that have read and understood the scientific literature, that have rationally compared and integrated all the scientific literature's conflicting statements, that can generate well founded novel hypotheses and theories, and that can suggest the most efficient experiments to advance science.
- There is a need to monitor and minimize the scientific misconduct that affects journals, reviewers, editors, and the scientific readership. Robust systems and independent auditing of research processes should be promoted to detect and prevent fraudulent or unethical practices and thus maintain scientific integrity.

5. AI Scientists

5.1. Background

The most advanced form of AI in science are 'AI Scientists' (also known as 'Robot Scientists', 'Self-driving Labs', 'Autonomous Discovery systems', 'Machine Scientists', etc.). These are AI systems integrated with laboratory automation to enable the closed-loop automation of scientific research. AI Scientists were first developed in Europe[68]. However, Europe trails well behind the US and Canada in investment in AI Scientists[69].

AI Scientists have multiple advantages. In terms of efficiency, AI Scientists are increasing the productivity of science. They can work cheaper, faster, more accurately, and longer than humans. They can also be easily multiplied. AI scientists also have the potential to ameliorate the 'Reproducibility Crisis', where in many areas of science, published results cannot be replicated in our laboratories. AI Scientists describe experiments in far greater detail and with greater semantic clarity than human scientists, and robots execute experimental protocols more accurately than humans. AI Scientists can be applied to almost all areas of research, including that in social science and humanities[70].

In April 2024, the European Commission's Scientific Advice Mechanism (SAM) released its independent policy recommendations on how to facilitate the uptake of AI in research and innovation across the EU. These policy recommendations were written by the European Commission's Group of Chief Scientific Advisors. The policy recommendation most relevant to AI Scientists is the "Establishment of a European Distributed Institute for AI in science". The recommendation states "to counter the dominance of a limited number of corporations over AI infrastructure and to empower public research across diverse disciplines, the scientists advise the creation of a new institute. This facility would offer extensive computational resources, a

[68] King, R.D., Whelan, K.E., Jones, F.M., Reiser, P.G.K., Bryant, C.H., Muggleton, S.H., Kell, D.B., & Oliver, S.G. (2004), Functional genomic hypothesis generation and experimentation by a robot scientist. *Nature*, 427, 247-252; King, R.D., Rowland, J., Oliver, S.G., Young, M., Aubrey, W., Byrne, E., Liakata, M., Markham, M., Pir, P., Soldatova, L.N., Sparkes, A., Whelan, K.E., Clare, C. (2009), The Automation of Science. Science. 324, 85-89; Williams K., Bilsland E., Sparkes A., Aubrey W., Young M., Soldatova L.N., De Grave K., Ramon J., de Clare M., Sirawaraporn W., Oliver S.G., & King R.D. (2015), Cheaper Faster Drug Development Validated by the Repositioning of Drugs Against Neglected Tropical Diseases. Royal Society Interface. DOI: 10.1098/rsif.2014.1289.

[69] University of Toronto (2024), https://www.utoronto.ca/news/experts-say-200-million-grant-awarded-u-t-will-drive-big-science-acceleration-consortium; National Institute of Standards and Technology (2024), NIST Special Publication NIST SP 1320, Driving U.S. Innovation in Materials and Manufacturing using AI and Autonomous Labs, https://doi.org/10.6028/NIST.SP.1320.

[70] Musslick, S., Bartlett, L.K., Chandramouli , S.H., Dubova, M.,Gobet, F., Griffiths, T., Slomanm, J., Hullman, J., King, R.D., Kutz, J.N., Lucas, C.G., Mahesh, S., Pestilli., F., Sloman, S.J., & Holmes, W.R. (2024), Automating the Practice of Science – Opportunities, Challenges, and Implications (2024) PNAS (in press).

sustainable cloud infrastructure and specialised AI trainings for scientists"[71]. This concept has informally been called 'A CERN for AI in Science' [72].
This concept is supported by the AI community in Europe, i.e. by both ELLIS and CAIRNE, as well as ADRA (AI, Data and Robotics Association)[73].

A CERN for AI in Science would give Europe the potential to dominate the next revolution in science and technology, and so safeguard its health, economic security, and military security of Europe. Such an Institute would be the world's premier site for AI in Science, the equivalent to CERN for particle physics. It would have access to world-class HPC and data resources, as well as the world's largest and most integrated collection of laboratory automation. It would have core responsibility in Europe for developing scientific Foundation Models, and AI scientists. A centralised site for laboratory automation at scale would drive the development of interoperability and AI readiness in the European laboratory automation industry, which is essential for the industry's future. One mechanism for furthering 'A CERN for AI in Science' could be the planned European AI Research Council[74].

Europe leads the world in laboratory automation; the industry is annually worth billions of euros. Of the World's top laboratory automation 50 suppliers, 17 are from Europe, 23 from the US and 10 from Japan. In addition, many of the units of the US-based companies are in Europe- therefore, R&D and production is mostly based in Europe. Leading European companies are: Tecan, Zeiss, Leica, Hamilton, Eppendorf, Qiagen, Sartorius, Roche, Merck, Bruker, Mettler Toledo, Anton Parr Oxford instruments, Biovia, and a long tail of innovative SMEs and start- ups. Most of these are based in South Germany and Switzerland. In some segments Europe clearly leads: Optical microscopy: European suppliers have more than 50% of the market (Zeiss and Leica); Liquid handling robotics: 75% of the supplier market; Bioprocessing: 80% of the market (Sartorius, Eppendorf, Merck); Blood analysis: about half of the market; Lab integration: half of the market; Solids handling: 80% of the market (Chemspeed, Mettler Toledo). Healthcare and biopharma research spend is predominately in the USA. So, this is a strong export market for European suppliers. There is therefore the possibility of Europe utilising its dominance in laboratory automation to lead the world in AI in science.

5.2. Survey Results

The concept of AI Scientists is still novel and not yet mainstream in Europe. While the survey findings cannot be generalised due to a very limited sample size, there was support among survey respondents for a substantial European AI Scientist resource: a hub-and-spoke model, with AI Scientist facilities spread around Europe in national sites, and a coordinating central site, was preferred. This would enable a sufficient mass of expertise to compete globally, facilitate the spread of expertise throughout the EU, and serve as an exemplar of how AI systems can enable collaboration at scale.

5.3. Pisa Meeting Results

Philip Piatkiewicz (Secretary General of ADRA) described Europe's position in the global race for technology, highlighting its competitive strengths and exploring how these can be leveraged through a coordinated approach. Europe is lagging behind other parts of the world in terms of both public and private investments in AI raising concerns about its ability to effectively address climate,

[71] European Commission (2024), 'Commission receives scientific advice on Artificial Intelligence uptake in research and innovation', Press Release, available at: https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/commission-receives-scientific-advice-artificial-intelligence-uptake-research-and-innovation-2024-04-15_en.
[72] Renda, A. (2024), 'Towards a European large-scale initiative on Artificial Intelligence', CEPS, available at: https://www.ceps.eu/ceps-publications/towards-a-european-large-scale-initiative-on-artificial-intelligence/.
[73] See CAIRNE, 2025: https://cairne.eu/cern-for-ai/
[74] See https://ec.europa.eu/commission/presscorner/detail/en%5E/ip_24_4621.

demographic, and other challenges it is likely to face in the coming years. He argued that Europe retains several competitive advantages:

- The tech giants over the coming decade will not be platforms like Google and Facebook. They will be based on frontier technologies which Europe has a good chance of developing and Generative AI could better support the European market, linguistical and cultural climate.
- Europe has areas of innovation where it is already world-beating. For example, public services such as health, energy and aerospace. These areas could be the basis for rising to global tech leadership over the coming decade.
- European tech ecosystems are becoming increasingly more connected and supporting the attraction and retention of talent.
- Europe is closing the overall funding gap, and from an investors point of view, the biggest societal challenges also represent some of the biggest market opportunities.

5.4. Key Challenges

The following key challenges were identified, which are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- **Lack of investment in AI Scientists:** Despite Europe's leadership in laboratory automation, investment in AI Scientists significantly lags behind competitors like the US and Canada[75]. Insufficient funding limits the development, deployment, and adoption of self-driving labs, creating barriers to achieving global competitiveness in automated science. A lack of investment by the European laboratory automation industry in integrating their AI products risks losing their competitive advantage. The cost of CERN is annually ~1$B, and it would seem that a similar level of funding would be required. Note that this amount is small compared to the investment into AI HPC[76] (see the 100$B investment in 'Stargate' announced by President Trump).
- **Validation, ethical considerations, and public trust:** The novel nature of AI Scientists introduces challenges in validation, reproducibility, and public trust. It is easy to imagine many non-ethical applications of AI Scientists, such as the development of biological and chemical weapons. As with any emerging technology, the public can easily develop misunderstandings and misperceptions about AI Scientists[77].
- **Talent pipeline and interdisciplinary expertise:** Developing AI Scientists requires expertise in AI, robotics, laboratory automation, and domain-specific sciences, yet Europe currently lacks academic training programmes to foster this interdisciplinary workforce.
- **Interoperability and Usability of Laboratory Systems:** Many current laboratory automation systems are not user-friendly or interoperable, posing significant challenges for scientists in effectively leveraging these tools. There are also robotics research challenges that arise when increasing levels of automation and when increasing the generality of tasks within the laboratory[78].

[75] USA Congress (2024), Bipartisan House Task Force on Artificial Intelligence.
[76] See the 100$B investment in 'Stargate' announced by President Trump: https://www.bbc.com/news/articles/cy4m84d2xz2o.
[77] Sartori, L., & Bocca, G. (2023), Minding the gap (s): public perceptions of AI and socio-technical imaginaries. *AI & Society*, *38*(2), 443-458.
[78] Angelopoulos, A., Cahoon, J. F., & Alterovitz, R. (2024), Transforming science labs into automated factories of discovery. *Science Robotics*, *9*(95), eadm6991.

5.5. Policy Recommendations

A number of key recommended actions for policymakers should be taken to address the key challenges highlighted above. These recommendations are based on information provided by the MLE country participants through the survey and during the Pisa meeting, in addition to desk research and the author's expert view.

- Funding a large-scale initiative for AI in Science should be considered, as well as the best way to structure such an initiative. There was a consensus among participating countries towards cooperation among Member States and more European cooperation.
- Enhancing interoperability of laboratory hardware and software will be essential to increase adoption and usability, particularly in multi-disciplinary and collaborative research environments. When publicly procuring laboratory automation systems, MLE countries and regions should therefore require that these systems are interoperable, and AI ready. This is essential for the future competitivity of the European Laboratory Automation Industry.
- MLE countries and regions currently lack academic training programmes to foster this interdisciplinary workforce. Addressing this skills gap necessitates targeted educational initiatives, the creation of new academic curricula, and incentives to attract global talent in the field.
- Europe should engage with and support international efforts to mitigate the dangers of the use of AI in science (e.g., the AI-Bio Forum[79]).
- There is a need to support robotics science to address current challenges in achieving higher levels of automation and enhancing the generality of tasks performed within laboratories.

[79] https://www.nti.org/news/nti-convenes-the-first-international-ai-bio-forum/.

6. Conclusion

The central conclusion of the report is that *Europe is still currently internationally competitive in AI in science, but must act quickly to retain its competitive advantage.*

The race for AI in science is central to the future prosperity of Europe, as technology based on science is crucial to the future of European industry, health, and security. The Big Tech companies in the US and China that currently dominate AI do not dominate AI in science, as science requires laboratories and equipment, which Big Tech companies have only limited access to. In contrast, European science has in place the necessary scientific expertise, and an extensive network of academic and industrial scientific laboratories and facilities. This human talent and equipment is orders of magnitude greater than that of Big Tech. Europe also has the advantage that it is home to much of the laboratory automation industry.

While Europe is still competitive in AI in science it must act quickly to retain its competitive advantage. US Big Tech companies have made no secret of their desire to dominate AI in science, in addition to the US Government, both of which are investing heavily in AI in science. Similarly, the Chinese Government wishes to dominate AI in science. Therefore, if Europe does not decisively support AI in science it is likely that in the medium-term Europe will no longer be competitive, the timeline for this is unclear as AI technology is advancing very quickly.

Only about half of the countries surveyed had specific policies in place to support AI in science. The countries lacking policies tended to be smaller countries. This MLE is therefore well-placed to provide guidance for countries that wish to initiate policies for AI in science.

No European country, not even the largest, can alone realistically hope to retain international competitiveness in AI in science. Therefore, there is a clear need for countries to collaborate and coordinate their policies.

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This thematic report comprises an analysis of national strategies and policies for fostering AI in science infrastructure, current challenges, and policy suggestions. It explores the state-of-the-art of the infrastructures for AI in science, examines the potential for innovations such as ‘AI Scientists’, and assesses Europe's competitiveness in this field. It is based on a survey of AI infrastructure across MLE participating countries, desk research, a country visit to Pisa which took place on 21 and 22 November 2024, and Ross D. King's expert views. It discusses high performance computing, data, and advanced infrastructure including Foundation Models and AI scientists.

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